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ORIGINAL RESEARCH

Associations of Urine Specific Gravity With Body Mass Index and Lean Body Mass at the Population Level: Implications for Hydration Monitoring

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Urine specific gravity (USG) thresholds are used in practice and research to determine hypohydration. However, some limited research has found that body size and body composition may impact USG, suggesting that fixed cutoffs may be insensitive. Cross-sectional data from 3,634 participants of the 2007–2008 National Health and Nutrition Examination Survey were analyzed. Along with USG, body mass index (BMI), estimated lean body mass (LBM), and dietary intake were quantified. Logistic regression models were used to evaluate whether higher quintiles of BMI and LBM were associated with elevated USG (USG ≥ 1.020 and ≥1.025) after accounting for dietary moisture and sodium. The USG (1.018 ± 0.0003 vs. 1.015 ± 0.0004); BMI (28.4 ± 0.2 vs. 28.0 ± 0.2 kg/m^2); LBM (60.9 ± 0.3 vs. 42.2 ± 0.2 kg); dietary moisture (3,401 ± 92 vs. 2,759 ± 49 g/day); and dietary sodium (4,171 ± 85 vs. 2,959 ± 50) were greater in men than in women ($p < .05$). Men and women in the fifth quintiles of BMI or LBM (vs. Quintile 1) had greater odds (2.00–3.68, $p < .05$) of elevated USG. (The only exception was for the association between BMI and USG ≥ 1.025 in men.) Being in Quintile 4 of LBM or BMI (vs. Quintile 1) also tended to be associated with higher odds of elevated of USG, though this pattern was more consistent when using USG ≥ 1.020 than USG ≥ 1.025. In summary, BMI and LBM are associated with USG at the population level. These results affirm that USG depends on body size and composition and raise questions about using fixed USG thresholds for determining hypohydration, particularly for people in the upper quintiles of BMI and LBM.

***Keywords**:* body composition, epidemiology, fluid, muscle mass, urinary metabolites

The collection and analysis of urine is frequently used to assess hydration status. Urine specific gravity (USG) is one such measure that compares the density of urine to that of distilled water, with higher values representing higher risks of hypohydration (Chadha et al., 2001). Beyond particle number, USG is influenced by the molecular mass of particles, meaning the presence of larger molecules can have a disproportionate effect on USG in comparison with a measure such as urine osmolality (Chadha et al., 2001). Although urine osmolality and USG are both used to assess hydration status, USG is a more practical method because of its relative low cost, ease of use, and portability (Casa et al., 2000). Indeed, in the National Athletic Trainers' Association's most recent position stand on fluid replacement for active individuals, the authors recommended using USG (from first morning voids) as a viable option to evaluate hydration status (McDermott et al., 2017).

An array of exercise and clinical research studies have used a USG threshold of ≥1.020 to define hypohydration (e.g., Carr et al., 2019; Heishman et al., 2021; Ylinenvaara et al., 2014). This threshold was also endorsed in the most recent joint position stand on nutrition and athletic performance from the American College of Sports Medicine, Academy of Nutrition and Dietetics, and Dietitians of Canada, though the authors acknowledged that a USG threshold of 1.025 should be considered in some cases to account for individual variability (Thomas et al., 2016). In addition, the National Collegiate Athletic Association uses a USG cutoff of <1.020 as an indicator of euhydration in its wresting weight management program (Cutrufello et al., 2021). Beyond its common use in athletes, a variety of population-based studies have relied on a USG threshold of 1.020 to define hypohydration/concentrated urine (e.g., Mao et al., 2021; Rosinger et al., 2019).

Despite its widespread use, the accuracy of USG for determining hypohydration has been questioned in certain circumstances. Factors that may impact USG include body size and muscle mass, which influence the amounts of urea, creatinine, and uric acid in the urine (Hamouti et al., 2010). These urinary metabolites arise from catabolic processes in muscle and therefore correlate positively with lean body mass (LBM) and muscle mass (Alvim et al., 2020; Baxmann et al., 2008; Heymsfield et al., 1983). This suggests that people with larger body sizes and LBM compartments would have naturally higher USG. In support of this idea, Hamouti et al. (2010) found that rugby players had greater muscle mass and USG than endurance runners. Furthermore, a moderate-sized correlation between muscle mass and USG ($r = .50$) was observed in their sample. By comparing data across investigations, it also appears that using USG ≥ 1.020 to define hypohydration leads to more false positives in larger athletes (e.g., wrestlers; Oppliger et al., 2005) than smaller athletes (e.g., endurance athletes; Popowski et al., 2001).

Some population-based investigations further corroborate that body size and composition could contribute to elevations in plasma and urine solutes. Using National Health and Nutrition Examination Survey (NHANES) data, Rosinger et al. (2016) reported that urine osmolality rose with increasing BMI categories, though they did not evaluate the effects of LBM. Ehlert and Wilson (2021) also analyzed NHANES data and reported that both BMI and LBM were associated with elevated urine osmolality. Likewise, Stookey

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et al. (2007) found that obesity may contribute to a state of plasma hypertonicity, which could have implications for urinary hydration monitoring. However, none of these population studies utilized USG.

Given that the literature on this topic has largely been conducted using small samples of athletes or has used urine osmolality, additional research with larger, more diverse samples is warranted. In 2007–2008, the NHANES included measurements of USG. The hypotheses of this secondary NHANES data analysis were that (a) USG would increase with LBM and body mass index (BMI) and that (b) higher odds of elevated USG would be observed with increasing levels of LBM and BMI.

Methods

Participants and Design

This cross-sectional, secondary analysis was carried out using the 2007–2008 NHANES survey cycle data. The NHANES is designed to be a representative source of health information on noninstitutionalized Americans of all ages from all 50 states and Washington DC (Johnson et al., 2013). Data collection consists of a combination of in-person home interviews as well as interviews and examinations at mobile exam centers (MECs). Oversampling of specific subgroups is done to increase the reliability and precision of estimates from those subgroups, and for 2007–2008, Hispanics, non-Hispanic Black persons, non-Hispanic White and other persons at or below 130% of the federal poverty level, and non-Hispanic White and other persons aged 80 and older were oversampled (Johnson et al., 2013).

The National Center for Health Statistics Ethics Review Board approved the NHANES protocols, and all adults gave written informed consent. All protocols followed the tenets of the Declaration of Helsinki. The protocol for this analysis was submitted to the human subjects review committee of the Darden College of Education and Professional Studies at Old Dominion University and was determined to be exempt from the review of the institutional review board.

A total of 12,943 individuals were screened, of which 10,149 participated in the household interviews and 9,762 participated in the MEC component. Individuals <20 years of age; pregnant women; and those with kidney disease, major cardiovascular conditions, and diabetes were excluded from this study. People without available data for relevant variables (BMI, estimated LBM, dietary intake, USG, race/ethnicity, and education) were also excluded. One individual was excluded for having a USG value that was an extreme outlier (1.20), while another was excluded for having an implausibly low sodium intake (0 mg/day). After accounting for these exclusions, 3,634 participants remained, representing 154.3 million Americans (Table 1).

Urine Specific Gravity

Urine samples were provided at MEC examinations. Participants were randomly assigned to a morning, afternoon, or evening appointment (Pfeiffer et al., 2017), meaning the urine samples were random spot specimens. The USG was measured using a hand-held refractometer that incorporates an automatic temperature compensation (Atago PAL-10S; Atago, Tokyo, Japan). A small sample of urine (0.3 ml) was placed on the prism surface, after which a reading was initiated and recorded. The refractometer was set to zero with water once for each run, and because there were 24 samples in each run, the refractometer was auto-zeroed every 24 samples (NHANES, 2011).

Table 1 Characteristics of the Sample by Gender

Variable	Men (n = 1,784)	Women (n = 1,850)
Age (years)	43.8 ± 0.6	45.4 ± 0.6
BMI (kg/m^2)	28.4 ± 0.2	28.0 ± 0.2
LBM (kg)	60.9 ± 0.3	42.2 ± 0.2
USG	1.018 ± 0.0003	1.015 ± 0.0004
Dietary moisture (g/day)	3,401 ± 92	2,759 ± 49
Dietary sodium (mg/day)	4,171 ± 85	2,959 ± 50
Education		
Ninth grade	6.0 ± 1.1%	4.8 ± 0.6%
Ninth to 11th grade	13.0 ± 1.3%	12.1 ± 1.4%
High school graduate or equivalent	26.0 ± 2.3%	24.8 ± 1.3%
Some college or AA degree	28.0 ± 1.1%	32.2 ± 1.2%
College graduate or above	27.0 ± 2.5%	26.0 ± 2.2%
Race/ethnicity		
Mexican American	9.9 ± 1.8%	7.3 ± 1.5%
Other Hispanic	5.2 ± 1.3%	4.6 ± 1.1%
Non-Hispanic White	69.0 ± 3.4%	72.8 ± 3.5%
Non-Hispanic Black	10.0 ± 1.6%	10.7 ± 2.3%
Other race	5.9 ± 1.2%	4.5 ± 1.0%

Note. AA = associate's degree; BMI = body mass index; LBM = lean body mass; USG = urine specific gravity.

Definition of Elevated USG

Participants were initially defined as having elevated USG if they had a USG of ≥1.020. However, in line with other investigations and recommendations (Savvides et al., 2020; Thomas et al., 2016), a USG threshold of ≥1.025 was used as a secondary definition of elevated USG. It is important to note that the goal was not to identify participants who were truly hypohydrated but to observe how the prevalence of being above these commonly used USG thresholds varied by BMI and LBM level.

Body Mass Index, Lean Body Mass, and Fat Mass

Body mass and height were measured by technicians with digital scales and stadiometers at the MEC and used to calculate BMI (in kilograms per meter squared). LBM was estimated based on equations that use demographic (age, gender, and race) and anthropometric (body mass, height, and waist circumference) variables (Lee et al., 2017).

- Men: 0.001 (age in years) + 0.064 (height in centimeters) + 0.756 (mass in kilograms) – 0.366 (waist circumference in centimeters) + 19.363 + (0 for White; –0.066 for Mexican American; 0.231 for Hispanic; 0.432 for Black; –1.007 for other race).
- Women: –0.039 (age in years) + 0.186 (height in centimeters) + 0.383 (mass in kilograms) – 0.043 (waist circumference in centimeters) – 10.683 + (0 for White; –0.359 for Mexican American; –0.059 for Hispanic; 1.085 for Black; –0.340 for other race).

As compared against dual-energy X-ray absorptiometry, these equations have R^2 values of .91 and .85 for men and women,

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respectively (Lee et al., 2017). For both BMI and LBM, data were split into quintiles for each gender to facilitate the analyses and interpretation of results.

Based on a reviewer's suggestion, fat mass (FM) was also calculated (body mass − LBM = FM) so that its association with USG could be evaluated. If LBM is more strongly associated with USG than FM, it provides further support for the relative greater importance of LBM on USG.

Additional Variables

Demographic information (age, gender, race/ethnicity, and education) was reported during household interviews. Pregnancy was determined for women through a combination of self-report and urine testing, and pregnant females or those whose status could not be ascertained were excluded.

The NHANES research has shown water intake to associate with urinary measures (Yeh et al., 2015) and increase with BMI (Kant et al., 2009). Hence, dietary moisture, which reflects moisture from all foods, beverages, and water consumed, was estimated from a 24-hr dietary recall (NHANES, 2013). Dietary sodium intake was also included in the analyses given its potential impact on USG (Chadha et al., 2001); dietary chloride was unavailable.

Given glucose's effects on USG (Voinescu et al., 2002), individuals who had a HbA_{1c} consistent with diabetes (≥6.5%) were excluded. Venous blood samples collected at the MECs were processed, stored, and shipped to Fairview Medical Center Laboratory at the University of Minnesota for analysis. HbA_{1c} measurements were derived from whole blood analysis on the A1c G7 high-performance liquid chromatographic Glycohemoglobin Analyzer (Tosoh Medics, Inc., San Francisco, CA). Individuals who reported weak/failing kidneys or cardiovascular conditions (congestive heart failure, heart attack, and stroke) were also excluded.

Statistical Analysis

Data files were downloaded from the NHANES website and processed with SAS (version 9.4; SAS Institute Inc., Cary, NC). To account for the complex design of NHANES and to generate nationally representative data, the analyses were carried out using the complex samples feature of SPSS (version 27.0; IBM Corp., Armonk, NY). Use of sample weights in the analyses accounted for oversampling of certain groups, nonresponse bias, and poststratification adjustments that are done to better reflect the U.S. population. Day one dietary sample weights were utilized in the present study. The National Center for Health Statistics recommends using Taylor Series Linearization methods for variance estimation (National Center for Health Statistics, n.d.), which is the approach used within the procedures of SPSS statistics complex samples.

Mean ± *SE* was calculated for continuous data, while prevalence estimates (*SE*) were calculated for categorical data (race/ethnicity, education, and presence of elevated USG). BMI, LBM, USG, dietary moisture, and dietary sodium all showed somewhat right-skewed distributions. However, using log-transformed data did not meaningfully impact the results, so untransformed data were used. Differences between men and women for selected continuous variables (USG, LBM, BMI, dietary moisture, and dietary sodium) were evaluated with a procedure in complex samples that is equivalent to a *t* test (IBM Support, 2020). Similarly, differences between quintile groups of BMI and LBM for selected continuous variables were evaluated using a procedure that is equivalent to one-way analysis of variance (IBM Support, 2020). Given dietary moisture and sodium intakes often correlate with BMI (Kant et al., 2009; Song et al., 2013), additional analyses included them as covariates, and estimated marginal means for USG were generated. Differences in categorical variables were assessed using the cross-tabs option and the test of independence of the row and column variables (adjusted Rao-Scott Pearson chi-square) (IBM, n.d).

Logistic regression within SPSS complex samples was used to assess whether higher BMI and LBM quintiles were associated with elevated USG as compared with reference groups (Quintile 1 for each variable). Dietary moisture and sodium were covariates. Odds ratios and 95% confidence intervals for the presence of elevated USG were calculated. Separate analyses were run for the two definitions of elevated USG, and analyses were done for men and women separately given gender differences in body composition and LBM. Finally, the strength of associations between body composition measures (LBM and FM) and USG were evaluated using general linear model regression. LBM and FM were entered into separate models predicting USG, and R^2 values were compared. All statistical tests with a $p < .05$ were considered significant.

Results

Descriptive statistics by gender are shown in Table 1. Men were slightly younger than women ($p = .019$), while mean USG (1.018 ± 0.0003 vs. 1.015 ± 0.0004; $p < .001$), LBM (60.9 ± 0.3 vs. 42.2 ± 0.2 kg; $p < .001$), and BMI (28.4 ± 0.2 vs. 28.0 ± 0.2 kg/m^2; $p = .022$) were all higher in men than women. In addition, dietary moisture and sodium intakes were greater in men ($p < .001$). The test of independence of the row and column variables was significant for race/ethnicity ($p = .018$) but not education level ($p = .12$).

The prevalence of elevated USG was significantly higher in men than in women when USG ≥ 1.020 was used (46.8 ± 2.0% vs. 27.6 ± 1.3%, $p < .001$). When USG ≥ 1.025 was used, men also had a higher prevalence of elevated USG (16.7 ± 1.6% vs. 11.5 ± 1.3%, $p = .004$).

Mean USG levels by quintiles of BMI and LBM are shown in Table 2 for men. In addition, estimated marginal means for USG are also shown for the analyses that included dietary moisture and sodium as covariates. For reference, mean BMI and LBM are shown for each quintile group. The same data are shown for women in Table 3. Both unadjusted and adjusted USG varied significantly across BMI and LBM quintile groups for men and women ($p < .05$), except for the analysis that looked at unadjusted USG and BMI in men ($p = .06$).

Figure 1a displays the prevalence of elevated USG among men by quintiles of BMI; Figure 1b replicates this except with LBM quintiles instead of BMI. Equivalent data for women are shown in Figure 2a and 2b. The test of independence of row and column variables was significant in all cases ($p < .05$), except when looking at the prevalence of USG ≥ 1.025 by BMI quintile in men ($p = .57$) and the prevalence of USG ≥ 1.025 by LBM quintile in men ($p = .30$).

The logistic regression models testing whether the odds of elevated USG increase with BMI and LBM quintiles are shown in Table 4. In none of the models were Quintiles 2 and 3 of BMI or LBM significantly associated with higher odds of elevated USG, with one exception (men in Quintile 3 of LBM had 1.82, 95% confidence interval [1.00, 3.30] the odds of USG ≥ 1.020 relative to men in LBM Quintile 1). Among the ORs that were statistically

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Table 2 The USG Values by Quintiles of BMI and LBM in Men ($n = 1{,}784$)

Variable	BMI Q1	BMI Q2	BMI Q3	BMI Q4	BMI Q5	*p* value
BMI (kg/m^2)	21.5 ± 0.16	25.1 ± 0.05	27.6 ± 0.04	30.2 ± 0.07	37.0 ± 0.33	
USG	1.017 ± 0.0006	1.017 ± 0.0006	1.018 ± 0.0006	1.019 ± 0.0006	1.019 ± 0.0004	.064
USG_{adj}	1.017 ± 0.0006	1.017 ± 0.0006	1.018 ± 0.0006	1.019 ± 0.0006	1.019 ± 0.0004	.034
Variable	**LBM Q1**	**LBM Q2**	**LBM Q3**	**LBM Q4**	**LBM Q5**	***p* value**
LBM (kg)	47.4 ± 0.3	53.5 ± 0.1	57.8 ± 0.1	63.1 ± 0.1	75.2 ± 0.6	
USG	1.016 ± 0.0006	1.018 ± 0.0006	1.018 ± 0.0005	1.019 ± 0.0005	1.019 ± 0.0005	.008
USG_{adj}	1.016 ± 0.0006	1.018 ± 0.0006	1.018 ± 0.0005	1.019 ± 0.0005	1.019 ± 0.0005	.003

Note. BMI = body mass index; LBM = lean body mass; Q1–Q5 = Quintiles 1–5; USG = urine specific gravity; USG_{adj} = urine specific gravity with dietary moisture and sodium as covariates.

Table 3 The USG Values by Quintiles of BMI and LBM in Women ($n = 1{,}850$)

Variable	BMI Q1	BMI Q2	BMI Q3	BMI Q4	BMI Q5	*p* value
BMI (kg/m^2)	20.7 ± 0.09	24.5 ± 0.06	27.6 ± 0.10	31.3 ± 0.10	39.6 ± 0.40	
USG	1.014 ± 0.0006	1.014 ± 0.0005	1.015 ± 0.0008	1.016 ± 0.0005	1.018 ± 0.0009	.002
USG_{adj}	1.014 ± 0.0005	1.014 ± 0.0005	1.015 ± 0.0008	1.016 ± 0.0005	1.018 ± 0.0009	.002
Variable	**LBM Q1**	**LBM Q2**	**LBM Q3**	**LBM Q4**	**LBM Q5**	***p* value**
LBM (kg)	33.3 ± 0.1	37.7 ± 0.1	40.7 ± 0.1	44.9 ± 0.1	53.9 ± 0.5	
USG	1.014 ± 0.0005	1.014 ± 0.0005	1.014 ± 0.0006	1.015 ± 0.0006	1.018 ± 0.0009	.002
USG_{adj}	1.014 ± 0.0005	1.014 ± 0.0005	1.014 ± 0.0006	1.016 ± 0.0006	1.018 ± 0.0008	<.001

Note. BMI = body mass index; LBM = lean body mass; Q1–Q5 = Quintiles 1–5; USG = urine specific gravity; USG_{adj} = urine specific gravity with dietary moisture and sodium as covariates.

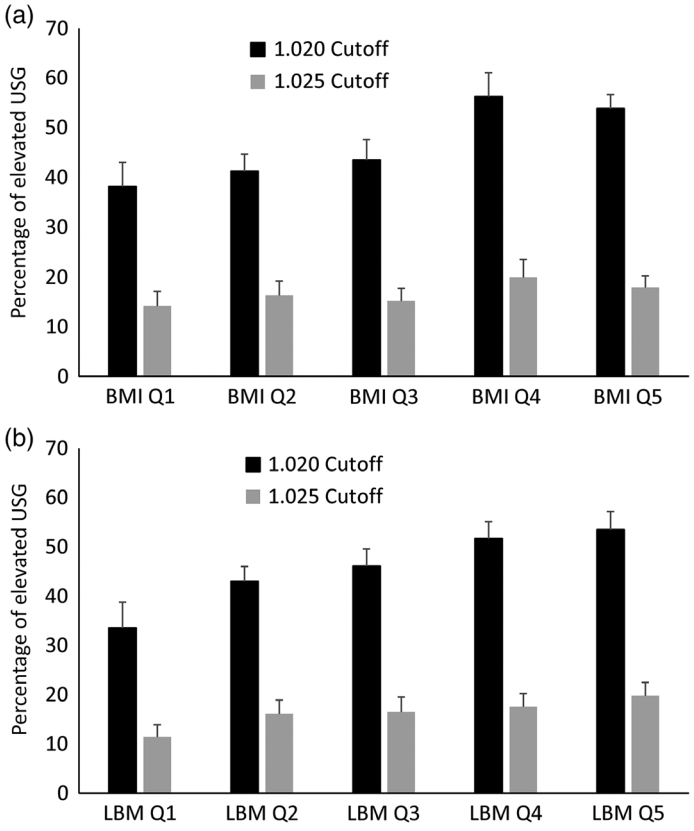

Figure 1 — Prevalence of elevated USG by quintiles of (a) BMI and (b) LBM in men. Error bars represent *SE*. USG = urine specific gravity; BMI = body mass index; LBM = lean body mass; Q1–Q5 = Quintiles 1–5.

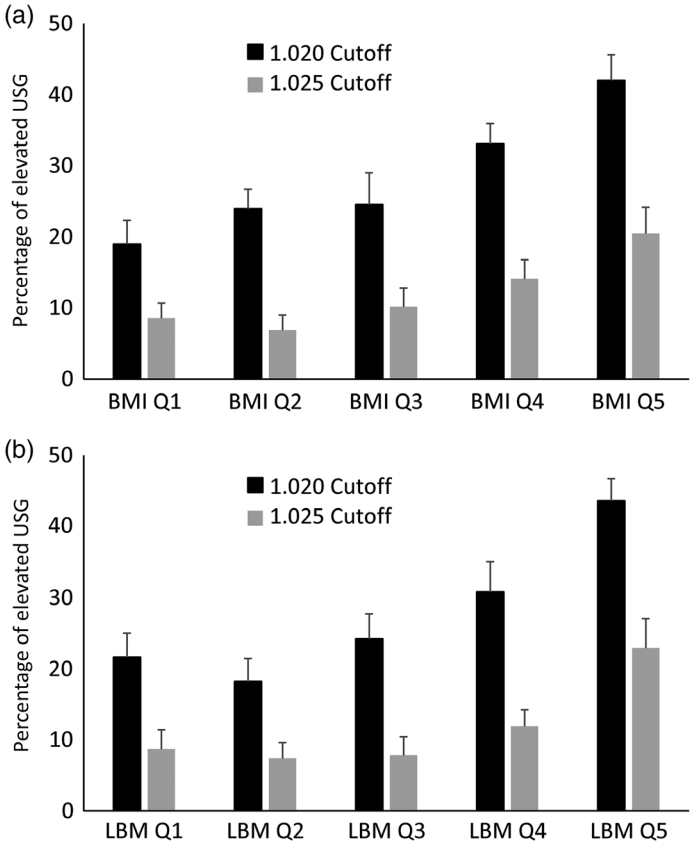

Figure 2 — Prevalence of elevated USG by quintile of (a) BMI and (b) LBM in women. Error bars represent *SE*. USG = urine specific gravity; BMI = body mass index; LBM = lean body mass; Q1–Q5 = Quintiles 1–5.

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Table 4 Odds of Elevated USG by Quintiles of BMI and LBM

Variable	Hypohydration (USG ≥ 1.020), OR [95% CI]	Hypohydration (USG ≥ 1.025), OR [95% CI]
Men		
BMI Q1	1.00 (reference)	1.00 (reference)
BMI Q2	1.17 [0.71, 1.93]	1.23 [0.72, 2.09]
BMI Q3	1.29 [0.78, 2.13]	1.16 [0.72, 1.86]
BMI Q4	2.20 [1.32, 3.68]*	1.63 [0.99, 2.66]
BMI Q5	2.00 [1.24, 3.24]*	1.40 [0.74, 2.66]
LBM Q1	1.00 (reference)	1.00 (reference)
LBM Q2	1.54 [0.94, 2.52]	1.54 [0.82, 2.88]
LBM Q3	1.82 [1.00, 3.30]*	1.66 [0.92, 3.00]
LBM Q4	2.30 [1.54, 3.45]*	1.82 [1.16, 2.86]*
LBM Q5	2.61 [1.43, 4.75]*	2.18 [1.28, 3.71]*
Women		
BMI Q1	1.00 (reference)	1.00 (reference)
BMI Q2	1.34 [0.75, 2.42]	0.79 [0.30, 2.05]
BMI Q3	1.43 [0.76, 2.69]	1.20 [0.52, 2.75]
BMI Q4	2.11 [1.16, 3.85]*	1.71 [0.79, 3.70]
BMI Q5	3.20 [1.73, 5.91]*	2.89 [1.18, 7.07]*
LBM Q1	1.00 (reference)	1.00 (reference)
LBM Q2	0.85 [0.47, 1.52]	0.89 [0.32, 2.46]
LBM Q3	1.24 [0.62, 2.47]	0.95 [0.34, 2.66]
LBM Q4	1.89 [1.07, 3.33]*	1.65 [0.69, 3.97]
LBM Q5	3.16 [1.87, 5.32]*	3.68 [1.27, 10.63]*

Note. The ORs are adjusted for dietary moisture and sodium intakes. BMI = body mass index; CI = confidence interval; LBM = lean body mass; OR = odds ratio; Q1–Q5 = Quintiles 1–5; USG = urine specific gravity; LBM = lean body mass. *$p < .05$.

significant, the sizes ranged from 1.82 to 3.68. Also, in comparison with BMI, LBM was consistently associated with modestly higher odds of elevated USG in men. In contrast, this same consistent pattern was not apparent in women.

Based on the general linear model regression analyses, LBM was, to a modest degree, more strongly associated with USG than FM in both men and women. The R^2 values for LBM and FM were .026 and .016 in men, respectively, while the R^2 values for LBM and FM were .060 and .043 in women, respectively. Furthermore, when including both LBM and FM as predictor variables into regression simultaneously, LBM (but not FM) remained statistically significantly associated ($p < .05$) with USG in both men and women.

Discussion

Consistent with studies of smaller and selective samples of athletes, this analysis of NHANES data found that USG is somewhat dependent on BMI and LBM. Furthermore, these associations are independent of dietary moisture and sodium intakes. As such, using fixed USG cutoffs to define hypohydration (≥1.020 and ≥1.025) may result in greater false positive rates of hypohydration for individuals who are in higher BMI and LBM groups. In this analysis, the associations of BMI and LBM with USG cutoffs were most pronounced among individuals in the fourth and fifth quintiles of these measures. Practically speaking, this means that most athletes would not be at markedly higher risk for elevated USG, except for athletes from sports in which large body sizes and high amounts of LBM are the norm (e.g., American football, rugby, sumo wrestling, weight throwing, and high weight class powerlifting). For nonathletes, however, the findings are likely to be relevant for a larger percentage of people given that approximately four in 10 adults have obesity (Hales et al., 2018).

From a mechanistic perspective, elevated USG among individuals with higher BMI and LBM is likely partly due to the presence of molecules in the urine that originate from muscle. Hamouti et al. (2010), for example, found that urine protein metabolites (urine uric acid + urea + creatinine) correlated very strongly with USG ($r = .92$, $p < .0001$), and these same metabolites showed a moderate correlation with muscle mass ($r = .47$, $p = .04$). Likewise, Alvim et al. (2020) demonstrated that children in the highest quartile of muscle mass had higher serum uric acid levels than those in lower quartiles of muscle mass. Beyond muscle-related metabolites, other substances in urine, such as electrolytes (particularly sodium and chloride), glucose, and so forth, impact USG (Chadha et al., 2001; Voinescu et al., 2002). Consequently, people with high sodium chloride intakes would be expected to have greater USG given that >90% of dietary sodium in excreted via urine (Lucko et al., 2018). Since people with elevated BMIs sometimes consume relatively more dietary sodium chloride (Song et al., 2013), the decision was made to include dietary sodium as a covariate. Ultimately, its inclusion had a negligible impact on the results.

The results of this analysis are in line with a previous NHANES study showing that urine osmolality is associated with BMI and LBM (Ehlert & Wilson, 2021). However, urine osmolality is typically not measured in field settings, preventing it from being a viable option for situations in which clinicians and researchers desire a convenient method of assessing hydration status. Consequently, researching the unique associations of BMI, LBM, and other body composition measures with USG should be prioritized given its widespread use. To date, relatively few studies have evaluated these associations, especially at the population level. One study of children aged 8–11 years found that there were no obvious differences in USG between overweight/obese and normal weight children, even though BMI was associated with urinary creatinine (Wang et al., 2015). The somewhat contrasting results with the present study could be due to their smaller sample size ($n = 243$) and/or the use of children who are different than adults in terms of body composition and kidney function (den Bakker et al., 2018).

This study has some notable limitations. The LBM was derived from predictive equations, and although this method shows strong agreement with dual-energy X-ray absorptiometry-derived LBM data, there is some additional error when predictive equations are used (Lee et al., 2017). It is possible, albeit speculative, that the associations between LBM and USG would have been stronger with more direct measures of LBM or muscle mass. Another limitation is that USG was determined from a single sample. Ideally, USG derived from a 24-hr period or longer would be used, but those data are often unavailable for large population studies. The spot samples were also collected at different times of the day, which, in terms of standardization, is not ideal but is more realistic when data from many subjects are collected. A morning sample would probably be preferable since it is more consistent from day-to-day, but USG tends to be elevated first thing in the morning due to lack of fluid intake overnight (Armstrong et al., 2010). In terms of strengths, this study's sample was large and

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diverse with respect to numerous factors (age, race/ethnicity, BMI, etc), making the results more generalizable than previous studies. In addition, the analyses accounted for dietary moisture and sodium intakes, which has not always been done in previous investigations.

In conclusion, the use of USG to assess hydration status is common in both clinical and field settings because of its availability and ease of measurement. However, some previous limited research has found that using fixed USG thresholds to identify hypohydration may be inappropriate across individuals who have highly variable levels of LBM. The present analysis bolsters these prior results by extending them to a large sample of American adults. Both BMI and LBM were associated with USG and the odds of elevated USG, particularly among those in the fourth and fifth quintiles of BMI and LBM. Further research is warranted to develop LBM- or BMI-specific USG thresholds for determining hypohydration status.

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The author downloaded and analyzed the data and wrote the entirety of the manuscript. The author has no conflicts of interest to declare.

References

Alvim, R.O., Siqueira, J.H., Zaniqueli, D., Dutra, D.M., Oliosa, P.R., & Mill, J.G. (2020). Influence of muscle mass on the serum uric acid levels in children and adolescents. *Nutrition, Metabolism and Cardiovascular Diseases, 30*(2), 300–305. https://doi.org/10.1016/j.numecd.2019.08.019

Armstrong, L.E., Pumerantz, A.C., Fiala, K.A., Roti, M.W., Kavouras, S.A., Casa, D.J., & Maresh, C.M. (2010). Human hydration indices: Acute and longitudinal reference values. *International Journal of Sport Nutrition and Exercise Metabolism, 20*(2), 145–153. https://doi.org/10.1123/ijsnem.20.2.145

Baxmann, A.C., Ahmed, M.S., Marques, N.C., Menon, V.B., Pereira, A.B., Kirsztajn, G.M., & Heilberg, I.P. (2008). Influence of muscle mass and physical activity on serum and urinary creatinine and serum cystatin C. *Clinical Journal of the American Society of Nephrology, 3*(2), 348–354. https://doi.org/10.2215/CJN.02870707

Carr, A., McGawley, K., Govus, A., Andersson, E.P., Shannon, O.M., Mattsson, S., & Melin, A. (2019). Nutritional intake in elite cross-country skiers during two days of training and competition. *International Journal of Sport Nutrition and Exercise Metabolism, 29*(3), 273–281. https://doi.org/10.1123/ijsnem.2017-0411

Casa, D.J., Armstrong, L.E., Hillman, S.K., Montain, S.J., Reiff, R.V., Rich, B.S., Roberts, W.O., & Stone, J.A. (2000). National athletic trainers' association position statement: Fluid replacement for athletes. *Journal of Athletic Training, 35*(2), 212–224.

Chadha, V., Garg, U., & Alon, U.S. (2001). Measurement of urinary concentration: A critical appraisal of methodologies. *Pediatric Nephrology, 16*(4), 374–382. https://doi.org/10.1007/s004670000551

Cutrufello, P.T., Landram, M.J., Venezia, A.C., & Dixon, C.B. (2021). A comparison of methods used to determine percent body fat, minimum wrestling weight, and lowest allowable weight class. *Journal of Strength & Conditioning Research, 35*(3), 633–637. https://doi.org/10.1519/JSC.0000000000003929

den Bakker, E., Gemke, R.J., & Bökenkamp, A. (2018). Endogenous markers for kidney function in children: A review. *Critical Reviews in Clinical Laboratory Sciences, 55*(3), 163–183. https://doi.org/10.1080/10408363.2018.1427041

Ehlert, A.M., & Wilson, P.B. (2021). The associations between body mass index, estimated lean body mass, and urinary hydration markers at the population level. *Measurement in Physical Education and Exercise Science, 25*(2), 163–170. https://doi.org/10.1080/1091367X.2020.1858300

Hales, C.M., Fryar, C.D., Carroll, M.D., Freedman, D.S., Aoki, Y., & Ogden, C.L. (2018). Differences in obesity prevalence by demographic characteristics and urbanization level among adults in the United States, 2013–2016. *JAMA, 319*(23), 2419–2429. https://doi.org/10.1001/jama.2018.7270

Hamouti, N., Del Coso, J., Ávila, A., & Mora-Rodriguez, R. (2010). Effects of athletes' muscle mass on urinary markers of hydration status. *European Journal of Applied Physiology, 109*(2), 213–219. https://doi.org/10.1007/s00421-009-1333-x

Heishman, A.D., Daub, B.D., Miller, R.M., Freitas, E.D., & Bemben, M.G. (2021). Longitudinal hydration assessment in collegiate basketball players over various training phases. *Journal of Strength & Conditioning Research, 35*(4), 1089–1094. https://doi.org/10.1519/JSC.0000000000002845

Heymsfield, S.B., Arteaga, C., McManus, C., Smith, J., & Moffitt, S. (1983). Measurement of muscle mass in humans: Validity of the 24-hour urinary creatinine method. *American journal of Clinical Nutrition, 37*(3), 478–494. https://doi.org/10.1093/ajcn/37.3.478

IBM Support. (2020). Retrieved from https://www.ibm.com/support/pages/how-do-i-do-independent-samples-t-test-or-anova-model-spss-statistics-complex-samples

IBM. (n.d.) Complex samples crosstabs. Retrieved from https://www.ibm.com/docs/en/spss-statistics/27.0.0?topic=samples-complex-crosstabs

Johnson, C.L., Paulose-Ram, R., Ogden, C.L., Carroll, M.D., Kruszan-Moran, D., Dohrmann, S.M., & Curtin, L.R. (2013). National health and nutrition examination survey: Analytic guidelines, 1999–2010. *Vital and Health Statistics. Series 2, Data Evaluation and Methods Research,* 161, 1–24. https://www.cdc.gov/nchs/data/series/sr_02/sr02_161.pdf

Kant, A.K., Graubard, B.I., & Atchison, E.A. (2009). Intakes of plain water, moisture in foods and beverages, and total water in the adult US population—Nutritional, meal pattern, and body weight correlates: National health and nutrition examination surveys 1999–2006. *American Journal of Clinical Nutrition, 90*(3), 655–663. https://doi.org/10.3945/ajcn.2009.27749

Lee, D.H., Keum, N., Hu, F.B., Orav, E.J., Rimm, E.B., Sun, Q., Willett W.C, & Giovannucci, E.L. (2017). Development and validation of anthropometric prediction equations for lean body mass, fat mass and percent fat in adults using the National Health and Nutrition Examination Survey (NHANES) 1999–2006. *British Journal of Nutrition, 118*(10), 858–866. https://doi.org/10.1017/S0007114517002665

Lucko, A.M., Doktorchik, C., Woodward, M., Cogswell, M., Neal, B., Rabi, D., Anderson, C., He, F.J., MacGregor, G.A., L'Abbe, M., Arcand, J., Whelton, P.K., McLean, R., Campbell, N.R.C., & TRUE Consortium. (2018). Percentage of ingested sodium excreted in 24-hour urine collections: A systematic review and meta-analysis. *Journal of Clinical Hypertension, 20*(9), 1220–1229. https://doi.org/10.1111/jch.13353

Mao, W., Zhang, H., Xu, Z., Geng, J., Zhang, Z., Wu, J., Xu, B., & Chen, M. (2021). Relationship between urine specific gravity and the prevalence rate of kidney stone. *Translational Andrology and Urology, 10*(1), 184–194. https://doi.org/10.21037/tau-20-929

McDermott, B.P., Anderson, S.A., Armstrong, L.E., Casa, D.J., Cheuvront, S.N., Cooper, L., Kenney, W.L, O'Connor, F.G, & Roberts, W.O. (2017). National athletic trainers' association position statement: Fluid replacement for the physically active. *Journal of Athletic Training, 52*(9), 877–895. https://doi.org/10.4085/1062-6050-52.9.02

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National Center for Health Statistics. (n.d.) Module 4: Variance estimation. Retrieved from https://wwwn.cdc.gov/nchs/nhanes/tutorials/module4.aspx

NHANES. (2011). 2007–2008 Data documentation, codebook, and frequencies. Urine specific gravity measurement (Surplus) (SSUSG_E). Retrieved from https://wwwn.cdc.gov/Nchs/Nhanes/2007-2008/SSUSG_E.htm

NHANES. (2013). 2007–2008 Data documentation, codebook, and frequencies. Dietary interview—Total nutrient intakes, first day (DR1TOT_E). Retrieved from https://wwwn.cdc.gov/Nchs/Nhanes/2007-2008/DR1TOT_E.htm

Oppliger, R.A., Magnes, S.A., Popowski, L.A., & Gisolfi, C.V. (2005). Accuracy of urine specific gravity and osmolality as indicators of hydration status. *International Journal of Sport Nutrition and Exercise Metabolism, 15*(3), 236–251. https://doi.org/10.1123/ijsnem.15.3.236

Pfeiffer, C.M., Lacher, D.A., Schleicher, R.L., Johnson, C.L., & Yetley, E.A. (2017). Challenges and lessons learned in generating and interpreting NHANES nutritional biomarker data. *Advances in Nutrition, 8*(2), 290–307. https://doi.org/10.3945/an.116.014076

Popowski, L.A., Oppliger, R.A., Lambert, P., Johnson, R.F., & Gisolf, C.V. (2001). Blood and urinary measures of hydration status during progressive acute dehydration. *Medicine & Science in Sports & Exercise, 33*(5), 747–753. https://doi.org/10.1097/00005768-200105000-00011

Rosinger, A.Y., Chang, A.M., Buxton, O.M., Li, J., Wu, S., & Gao, X. (2019). Short sleep duration is associated with inadequate hydration: Cross-cultural evidence from US and Chinese adults. *Sleep, 42*(2), zsy210. https://doi.org/10.1093/sleep/zsy210

Rosinger, A.Y., Lawman, H.G., Akinbami, L.J., & Ogden, C.L. (2016). The role of obesity in the relation between total water intake and urine osmolality in US adults, 2009–2012. *American Journal of Clinical Nutrition, 104*(6), 1554–1561. https://doi.org/10.3945/ajcn.116.137414

Savvides, A., Giannaki, C.D, Vlahoyiannis, A., Stavrinou, P.S., & Aphamis, G. (2020). Effects of dehydration on archery performance, subjective feelings and heart rate during a competition simulation. *Journal of Functional Morphology and Kinesiology, 5*(3), 67. https://doi.org/10.3390/jfmk5030067

Song, H.J., Cho, Y.G., & Lee, H.J. (2013). Dietary sodium intake and prevalence of overweight in adults. *Metabolism, 62*(5), 703–708. https://doi.org/10.1016/j.metabol.2012.11.009

Stookey, J.D., Barclay, D., Arieff, A., & Popkin, B.M. (2007). The altered fluid distribution in obesity may reflect plasma hypertonicity. *European Journal of Clinical Nutrition, 61*(2), 190–199. https://doi.org/10.1038/sj.ejcn.1602521

Thomas, D.T., Erdman, K.A., & Burke, L.M. (2016). American college of sports medicine joint position statement. Nutrition and athletic performance. *Medicine & Science in Sports & Exercise, 48*(3), 543–568. https://doi.org/10.1249/MSS.0000000000000852

Voinescu, G.C., Shoemaker, M., Moore, H., Khanna, R., & Nolph, K.D. (2002). The relationship between urine osmolality and specific gravity. *American Journal of the Medical Sciences, 323*(1), 39–42. https://doi.org/10.1097/00000441-200201000-00007

Wang, B., Tang, C., Wang, H., Zhou, W., Chen, Y., Zhou, Y., & Jiang, Q. (2015). Influence of body mass index status on urinary creatinine and specific gravity for epidemiological study of children. *European Journal of Pediatrics, 174*(11), 1481–1489. https://doi.org/10.1007/s00431-015-2558-9

Yeh, H.C., Lin, Y.S., Kuo, C.C., Weidemann, D., Weaver, V., Fadrowski, J., Neu, A., & Navas-Acien, A. (2015). Urine osmolality in the US population: Implications for environmental biomonitoring. *Environmental Research, 136,* 482–490. https://doi.org/10.1016/j.envres.2014.09.009

Ylinenvaara, S.I., Elisson, O., Berg, K., Zdolsek, J.H., Krook, H., & Hahn, R.G. (2014). Preoperative urine-specific gravity and the incidence of complications after hip fracture surgery: A prospective, observational study. *European Journal of Anaesthesiology, 31*(2), 85–90. https://doi.org/10.1097/01.EJA.0000435057.72303.0e

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